

ANSWER SCHEME (MUST 'A' MATHS)

NO.	SOLUTION	MARKS	REMARKS
1(a)	$\lim_{x \rightarrow 3} \left(\frac{\sqrt{x-1}-\sqrt{2}}{x-3} \right) \left(\frac{\sqrt{x-1}+\sqrt{2}}{\sqrt{x-1}+\sqrt{2}} \right)$ $= \lim_{x \rightarrow 3} \left(\frac{x-3}{(x-3)(\sqrt{x-1}+\sqrt{2})} \right)$ $= \lim_{x \rightarrow 3} \left(\frac{1}{\sqrt{x-1}+\sqrt{2}} \right) = \frac{1}{2\sqrt{2}}$	K1 K1J1	Multiply with conjugate Simplify and substitute $x = 3$
(b)	$\lim_{x \rightarrow -\infty} \left(\frac{\sqrt{x^2+5}}{2-x} \right) = \lim_{x \rightarrow -\infty} \frac{-\sqrt{\frac{x^2}{x^2} + \frac{5}{x^2}}}{\frac{2}{x} - \frac{x}{x}}$ $= \lim_{x \rightarrow -\infty} \frac{-\sqrt{1 + \frac{5}{x^2}}}{\frac{2}{x} - 1}$ $= \lim_{x \rightarrow -\infty} \frac{-\sqrt{1+0}}{0-1} = 1$	K1 K1J1	Divide by x both numerator and denominator including negative sign Simplify and substitute
(c)	$\lim_{x \rightarrow 2^-} \frac{ x^2+x-6 }{2x-x^2} = \lim_{x \rightarrow 2^-} \frac{ (x+3)(x-2) }{x(2-x)}$ $= \lim_{x \rightarrow 2^-} \frac{ x+3 x-2 }{x(2-x)}$ $= \lim_{x \rightarrow 2^-} \frac{-(x-2) x+3 }{-x(x-2)}$ $= \lim_{x \rightarrow 2^-} \frac{ x+3 }{x} = \frac{5}{2}$	K1 K1J1	Factorise and choose the correct function Simplify and substitute $x = 2$
Total		9 marks	

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2	$y = xe^2 - \ln(2x+1)^{x^2}$ $= e^2x - x^2 \ln(2x+1)$ Let $u = x^2$, $u' = 2x$ $v = \ln(2x+1), \quad v' = \frac{2}{2x+1}$ $\frac{dy}{dx} = e^2 - \left[x^2 \left(\frac{2}{2x+1} \right) + 2x \ln(2x+1) \right]$ $= e^2 - \frac{2x^2}{2x+1} - 2x \ln(2x+1)$	B1 K1 K1K1 J1	Using properties of logarithm Let u, v and differentiate correctly K1-Differentiate xe^2 K1-substitute into correct formula
	$y = x^3 \sin^2 3x$ Let $u = x^3$, $u' = 3x^2$ $v = (\sin 3x)^2, \quad v' = 6 \sin 3x \cos 3x$ $\frac{dy}{dx} = x^3 (6 \sin 3x \cos 3x) + (\sin^2 3x)(3x^2)$ $= 6x^3 \sin 3x \cos 3x + 3x^2 \sin^2 3x$ $= 3x^2 \sin 3x (2x \cos 3x + \sin 3x)$	K1J1 K1 J1	Let u, v and differentiate correctly Substitute and use correct formula
Total		9 marks	

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NO.	SOLUTION	MARKS	REMARKS												
3	$f(x) = x^4 - 2x^3 + 10$ $f'(x) = 4x^3 - 6x^2$ $f''(x) = 12x^2 - 12x$ $f''(x) = 12x(x-1) = 0$ $x = 0, x = 1$ <table border="1"> <tr> <td>Interval</td><td>$(-\infty, 0)$</td><td>$(0, 1)$</td><td>$(1, \infty)$</td></tr> <tr> <td>Sign of $\frac{d^2y}{dx^2}$</td><td>+</td><td>-</td><td>+</td></tr> <tr> <td>conclusion</td><td>Concave upward</td><td>Concave downward</td><td>Concave upward</td></tr> </table> $f(0) = 0^4 - 2(0)^3 + 10 = 10$ $f(1) = 1^4 - 2(1)^3 + 10 = 9$ Since $f''(x)$ changes sign at $x = 0$ and $x = 1$, the points of inflection are $(0, 10)$ and $(1, 9)$	Interval	$(-\infty, 0)$	$(0, 1)$	$(1, \infty)$	Sign of $\frac{d^2y}{dx^2}$	+	-	+	conclusion	Concave upward	Concave downward	Concave upward	K1 K1 J1 K1 K1 J1 J1	Find the first derivative Find the second derivative and value of x when $f''(x) = 0$ Table of sign Find the values of y
Interval	$(-\infty, 0)$	$(0, 1)$	$(1, \infty)$												
Sign of $\frac{d^2y}{dx^2}$	+	-	+												
conclusion	Concave upward	Concave downward	Concave upward												
Total		7 marks													

ANSWER SCHEME (MUST 'A' MATHS)

SECTION B

NO.	SOLUTION	MARKS	REMARKS
1 (a)	$w = \frac{1}{\sqrt{3} + \sqrt{-1}}$ $= \frac{1}{(\sqrt{3} + i)(\sqrt{3} - i)}$ $= \frac{\sqrt{3} - i}{3 - i^2} = \frac{\sqrt{3} - i}{4}$ $= \frac{\sqrt{3}}{4} - \frac{1}{4}i$	K1	Find w in the form of $a + bi$
	$ w = \left \frac{\sqrt{3}}{4} - \frac{1}{4}i \right = \sqrt{\left(\frac{\sqrt{3}}{4} \right)^2 + \left(\frac{-1}{4} \right)^2}$ $= \sqrt{\frac{4}{16}} = \frac{1}{2}$	K1	Find modulus of w
		J1	Answer
	(b) the argument of $w, \theta = -\tan^{-1}\left(\frac{1/4}{\sqrt{3}/4}\right)$	K1	Find the argument of w
	$\theta = -\tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = -\frac{\pi}{6} \text{ radian}$	J1	Answer
	$\therefore w = \frac{1}{2} \left(\cos\left(-\frac{\pi}{6}\right) + i \sin\left(-\frac{\pi}{6}\right) \right)$	B1	Polar form of w
Total		6 marks	

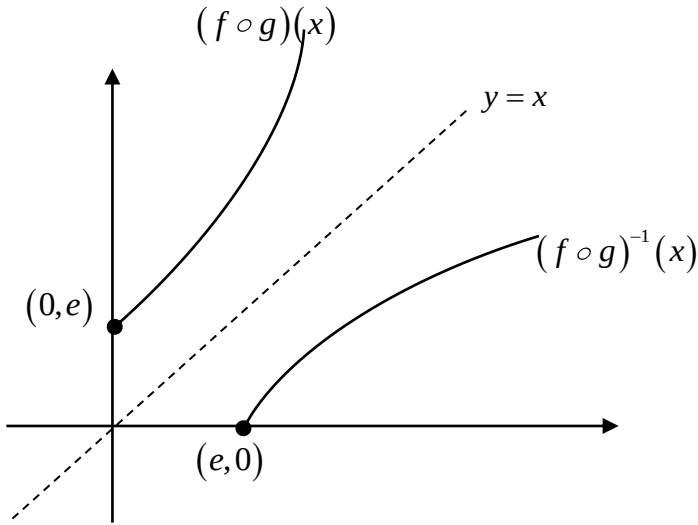
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NO.	SOLUTION	MARKS	REMARKS																
2	(a)																		
		$\ln x + 12 \log_x e = 7$																	
		$\ln x + 12 \frac{\log_e e}{\ln x} = 7$	K1	Change to the base e															
		$(\ln x)^2 + 12 = 7 \ln x$																	
		$(\ln x)^2 - 7 \ln x + 12 = 0$	K1	Quadratic form															
		$(\ln x - 4)(\ln x - 3) = 0$																	
		$\ln x - 4 = 0 \rightarrow x = e^4$	K1	Solve for $\ln$															
	$\ln x - 3 = 0 \rightarrow x = e^3$	J1	Answer																
	(b)																		
		$\frac{x^2 + 5}{x - 2} - x \leq 0$																	
		$\frac{x^2 + 5 - x(x - 2)}{x - 2} \leq 0$	K1	Make as single fraction															
		$\frac{5 + 2x}{x - 2} \leq 0$	J1	Correct rational inequality															
		<table><tr><td>Interval</td><td>$\left(-\infty, -\frac{5}{2}\right]$</td><td>$\left[-\frac{5}{2}, 2\right)$</td><td>$(2, +\infty]$</td></tr><tr><td>$5 + 2x$</td><td>–</td><td>+</td><td>+</td></tr><tr><td>$x - 2$</td><td>–</td><td>–</td><td>+</td></tr><tr><td>$\frac{5 + 2x}{x - 2}$</td><td>+</td><td>–</td><td>+</td></tr></table>	Interval	$\left(-\infty, -\frac{5}{2}\right]$	$\left[-\frac{5}{2}, 2\right)$	$(2, +\infty]$	$5 + 2x$	–	+	+	$x - 2$	–	–	+	$\frac{5 + 2x}{x - 2}$	+	–	+	K1
Interval		$\left(-\infty, -\frac{5}{2}\right]$	$\left[-\frac{5}{2}, 2\right)$	$(2, +\infty]$															
$5 + 2x$		–	+	+															
$x - 2$		–	–	+															
$\frac{5 + 2x}{x - 2}$	+	–	+																
$\therefore \left[-\frac{5}{2}, 2\right)$	K1	Sign																	
	J1	Answer																	
Total		9 marks																	

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3	$h^2(x) = h\left(\frac{2x}{x-2}\right) = \frac{2\left(\frac{2x}{x-2}\right)}{\left(\frac{2x}{x-2}\right) - 2}$	K1	Find the composite function
	$= \frac{\frac{4x}{x-2}}{\frac{2x-2x+4}{x-2}} = \frac{\frac{4x}{x-2}}{\frac{4}{x-2}}$	K1	Simplify
	$= x$	J1	Answer
	since $h^2(x) = x$, $\therefore h^{-1}(x) = h(x)$	K1 J1	Deduction and conclusion
	$h^7(8) = h(8) = \frac{2(8)}{8-2} = \frac{8}{3}$	K1 J1	Find the composite
Total		7 marks	
4	$f: x \rightarrow e^{x+1}$ and $g: x \rightarrow x $.		
	$f \circ g(x) = f(x) = e^{ x +1}$ (shown)	K1 J1	Find the composite function
	(a) let $(f \circ g)(x_1) = (f \circ g)(x_2)$ $e^{ x_1 +1} = e^{ x_2 +1}$ $ x_1 +1 = x_2 +1$ $ x_1 = x_2 $ $x_1 = \pm x_2$	K1	Algebraic approach
	since $f \circ g(x)$ is not a one-to-one function, $(f \circ g)^{-1}(x)$ does not exist.	J1	Answer
		J1	Conclusion

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NO.	SOLUTION	MARKS	REMARKS
(b)	$f \circ g(x) = \begin{cases} e^{x+1}, & x \geq 0 \\ e^{-x+1}, & x < 0 \end{cases}$	K1 K1 J1	Correct domain Correct $(f \circ g)(x)$
(c)(i)	Domain of $f \circ g(x)$ to have inverse: $x \geq 0$ or $x \leq 0$. $\therefore k = 0$ $(f \circ g)(f \circ g)^{-1}(x) = x$ $e^{(f \circ g)^{-1}(x)+1} = x$ $(f \circ g)^{-1}(x) = \ln x - 1$	B1 K1 J1	Value of k Find $(f \circ g)^{-1}(x)$ Answer
(c)(ii)		 R1 R1 R1	 Graph shape for both Any graph correct All correct
Total		14 marks	

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NO.	SOLUTION	MARKS	REMARKS
5(a)	$D_f : \frac{x+1}{x-1} \geq 0 \Rightarrow D_f = (-\infty, -1] \cup (1, \infty)$		
	$\lim_{x \rightarrow 1^+} \sqrt{\frac{x+1}{x-1}} = +\infty$ $\therefore x=1$ is a vertical asymptote	K1 J1	Method to show vertical asymptote
	$\lim_{x \rightarrow \pm \infty} \sqrt{\frac{x+1}{x-1}} = \lim_{x \rightarrow \pm \infty} \sqrt{\frac{\frac{x}{x} + \frac{1}{x}}{\frac{x}{x} - \frac{1}{x}}}$	K1	Method to show horizontal asymptote
	$= \sqrt{\frac{1+0}{1-0}} = 1$ $\therefore y=1$ is a horizontal asymptote	K1 J1	$\lim_{x \rightarrow \infty} \frac{1}{x} = 0$
5(b)	Given f is continuous on $(0,3)$, therefore f is continuous only at $x=2$	B1	
	$f(2) = \lim_{x \rightarrow 2} f(x)$	K1	
	$\lim_{x \rightarrow 2^+} ax^2 = \lim_{x \rightarrow 2^-} \frac{x}{x} + a$	K1K1	Correct function from left and right
	$4a = 1 + a$	K1	Simplify
	$3a = 1 \Rightarrow a = \frac{1}{3}$	J1	
	$f(5) = 3$	K1	Substitute in correct function
	$1 - b(5) = 3$	K1	Simplify
	$1 - 5b = 3$		
	$5b = -2 \Rightarrow b = \frac{-2}{5}$	J1	
Total		14 marks	

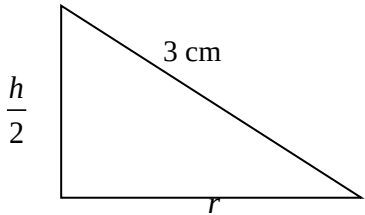
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NO.	SOLUTION	MARKS	REMARKS
6(a)	$f'(2^-) = \lim_{x \rightarrow 2^-} \frac{\frac{4}{3}x^2 - \frac{16}{3}}{x-2}$	K1	Substitute into correct formula
	$= \lim_{x \rightarrow 2^-} \frac{4(x-2)(x+2)}{3(x-2)}$	K1	Simplify and attempt to solve
	$= \lim_{x \rightarrow 2^-} \frac{4}{3}(x+2)$	J1	
	$= \frac{16}{3}$		
	$f'(2^+) = \lim_{x \rightarrow 2^+} \frac{2x + \frac{4}{3} - \frac{16}{3}}{x-2}$	K1	Substitute into correct formula
	$= \lim_{x \rightarrow 2^+} \frac{2x-4}{x-2}$		
	$= \lim_{x \rightarrow 2^+} \frac{2(x-2)}{(x-2)}$	K1	Simplify and attempt to solve
	$= \lim_{x \rightarrow 2^+} 2$	J1	
	$= 2$		
	Since $f'(2^-) \neq f'(2^+)$, $f'(2)$ does not exist.	J1	
6(b)	$x = \frac{1}{1-t^2} = (1-t^2)^{-1}$		
	$\frac{dx}{dt} = -(1-t^2)^{-2}(-2t)$	K1	Find $\frac{dx}{dt}$
	$= \frac{2t}{(1-t^2)^2}$	J1	
	$\frac{dt}{dx} = \frac{(1-t^2)^2}{2t}$		

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	$y = \frac{1+t^2}{t}$ $\frac{dy}{dt} = \frac{t(2t) - (1+t^2)}{t^2}$ $= \frac{t^2 - 1}{t^2}$	K1 J1	Find $\frac{dy}{dt}$
	$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$ $= \frac{(t^2 - 1)}{t^2} \cdot \frac{(1-t^2)^2}{2t}$ $= \frac{-(1-t^2)^3}{2t^3}$ $= -\frac{1}{2} \left(\frac{1-t^2}{t} \right)^3 \quad (\text{shown})$	K1 J1	Substitute into chain rule
	when $x = -\frac{1}{8}$, $-\frac{1}{8} = \frac{1}{1-t^2}$ $1-t^2 = -8$ $t^2 = 9$ $t = \pm 3$	K1	Find t

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	when $t = -3, y = \frac{1+(-3)^2}{-3} = \frac{10}{-3}$ (not valid) when $t = 3, y = \frac{1+(3)^2}{3} = \frac{10}{3}$ (valid) $\left. \frac{dy}{dx} \right _{t=3} = -\frac{1}{2} \left(\frac{1-3^2}{3} \right)^3 = \frac{256}{27}$	K1J1	
Total		16 marks	
7 (a)	 $r^2 = 3^2 - \left(\frac{h}{2}\right)^2$ $r^2 = 9 - \frac{h^2}{4}$ $V = \pi r^2 h$ $V = \pi h \left(9 - \frac{h^2}{4} \right)$	K1	Express r^2 in terms of h

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7(b)	$V = \pi h \left(9 - \frac{h^2}{4} \right) = \pi \left(9h - \frac{h^3}{4} \right)$	J1	As V in terms of h
	$\frac{dV}{dh} = \pi \left(9 - \frac{3h^2}{4} \right) = 0$	K1	Correct differentiation of $\frac{dV}{dh}$
	$h = \pm 2\sqrt{3}$		
	$h = 2\sqrt{3}$ since $h > 0$	K1	Find h when $\frac{dV}{dh} = 0$
	$\frac{d^2V}{dh^2} = \pi \left(-\frac{3}{2}h \right) = -3\sqrt{3}\pi < 0$	K1	Find second derivative and substitute value of h
	Therefore, V has maximum value when $h = 2\sqrt{3}$	J1	As it is
	$\frac{dV}{dh} = \pi \left(9 - \frac{3h^2}{4} \right)$		
	$\frac{dV}{dh} = \pi \left(9 - \frac{3(2)^2}{4} \right)$	K1	Substitute $h = 2$ into $\frac{dV}{dh}$
	$= 6\pi$		
	$\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$		
	$= \frac{1}{6\pi} \times (48)$	K1	Use chain rule
	$= \frac{8}{\pi} \text{cms}^{-1}$	J1	
Total		9 marks	
Total		75 Marks	